

Data Analysis Project 2.

Textual Analysis using Large Language Models (LLMs)

1. Project Overview

The goal of this project is to use Large Language Models (LLMs) to analyze textual data and generate inferences about firm outcomes (e.g., stock returns, earnings surprises, credit risk, ESG performance, litigation risk, innovation output). You will apply state-of-the-art AI tools to explore the explanatory power of novel data sources in helping investors make better decisions. While prior coursework introduced structured data and traditional econometric modeling, this project emphasizes unstructured textual data and modern NLP techniques. Unlike Project 1, you have full flexibility in topic, dataset, and methodology—as long as your work leverages LLMs and is grounded in an economic or financial framework.

*Students are encouraged to use AI tools for coding and data collection. All routine/codifiable tasks are allowed to be delegated to AI/AI agents to minimize human efforts.

2. Grading and Submission (at the group-level):

- Due by the presentation day (week 9)
- 20% credits will be granted based on the completion and degree of perfection.
- Submit the program (code) used for basic data collection and analysis in a zip file.
- Submit a report summary (see next page for detailed requirements)

3. Step-by-Step Guidelines

Choose Your Research Question - link textual data to a measurable economic outcome.

Examples:

- Does CEO tone in earnings calls predict firm performance?
- Can some features extracted by LLMs from patent descriptions forecast future technological breakthroughs?
- How does social media sentiment relate to stock return or volatility?
- Do LLM-based ESG scores predict changes in stock trading activities?

Select a Textual Dataset

Examples:

- Corporate disclosures: financial statements, earnings call transcripts (e.g., SEC Edgar, Seeking Alpha)
- News articles: Dow Jones, Bloomberg, Factiva
- Social media posts: Twitter/X, Reddit, StockTwits
- Patent filings: USPTO

*Any textual disclosure from Chinese-listed public firms and government websites will also be acceptable. You may also use self-constructed datasets scraped from the web (ethically and legally). Given the limited access to textual data, there is no requirement for the sample size.

*** Following datasets are available upon request (contact me if you need them):**

1. Earnings call transcripts (global coverage)
2. US firms' patent descriptions
3. US firms' announcement of mergers and acquisitions
4. Official task descriptions required by each job occupation
5. US firms' annual report (https://sraf.nd.edu/sec-edgar-data/lm_10x_summaries/)

Preprocessing & Data Preparation (select one approach based on your preference)

- Embedding generation (using open-source LLMs like LLaMA, Mistral, or API-based models)
- Summarization or topic modeling to extract key information
- Sentiment scoring (e.g., finance-specific lexicons, fine-tuned LLM sentiment classification)
- Any other LLM Application on classification through prompt engineering

**Example approaches:*

- Sentiment Analysis: Classify tone, optimism, or uncertainty in text.
- Topic Modeling / Clustering: Identify recurring themes in corporate communication.
- Information Extraction: Detect mentions of risk factors, competitive advantages, or ESG concerns.
- Custom Prompting / Zero-Shot Learning: Use LLMs to answer domain-specific queries directly from text.
- Embedding Similarity Analysis: Compare semantic distances between firm disclosures across time.

***Suggested tutorials for model selection and coding:**

<https://huggingface.co/models>

<https://dmxapi.cn/models.html#code-block>

https://colab.research.google.com/github/googlecolab/colabtools/blob/main/notebooks/Getting_started_with_google_colab_ai.ipynb#scrollTo=Ucchuu5vV3Jp

Economic / Financial Framework – link the analysis to theories

***The project must relate findings to theory or empirical literature, such as:**

- Market efficiency
- Information asymmetry and disclosure theory
- Behavioral finance and investor sentiment
- Risk factor models and asset pricing
- Innovation and growth theories

Empirical Testing - link extracted textual features to firm outcomes

Examples:

- Correlation test
- Simple Regression models, panel data methods, or machine learning prediction models

Written Report (max 20 pages) will include:

- Data Description & Preprocessing Steps
- Methodology (LLM architecture, prompts, parameter settings, training/fine-tuning if applicable)
- Empirical Results & Interpretation
- Conclusion & Limitations